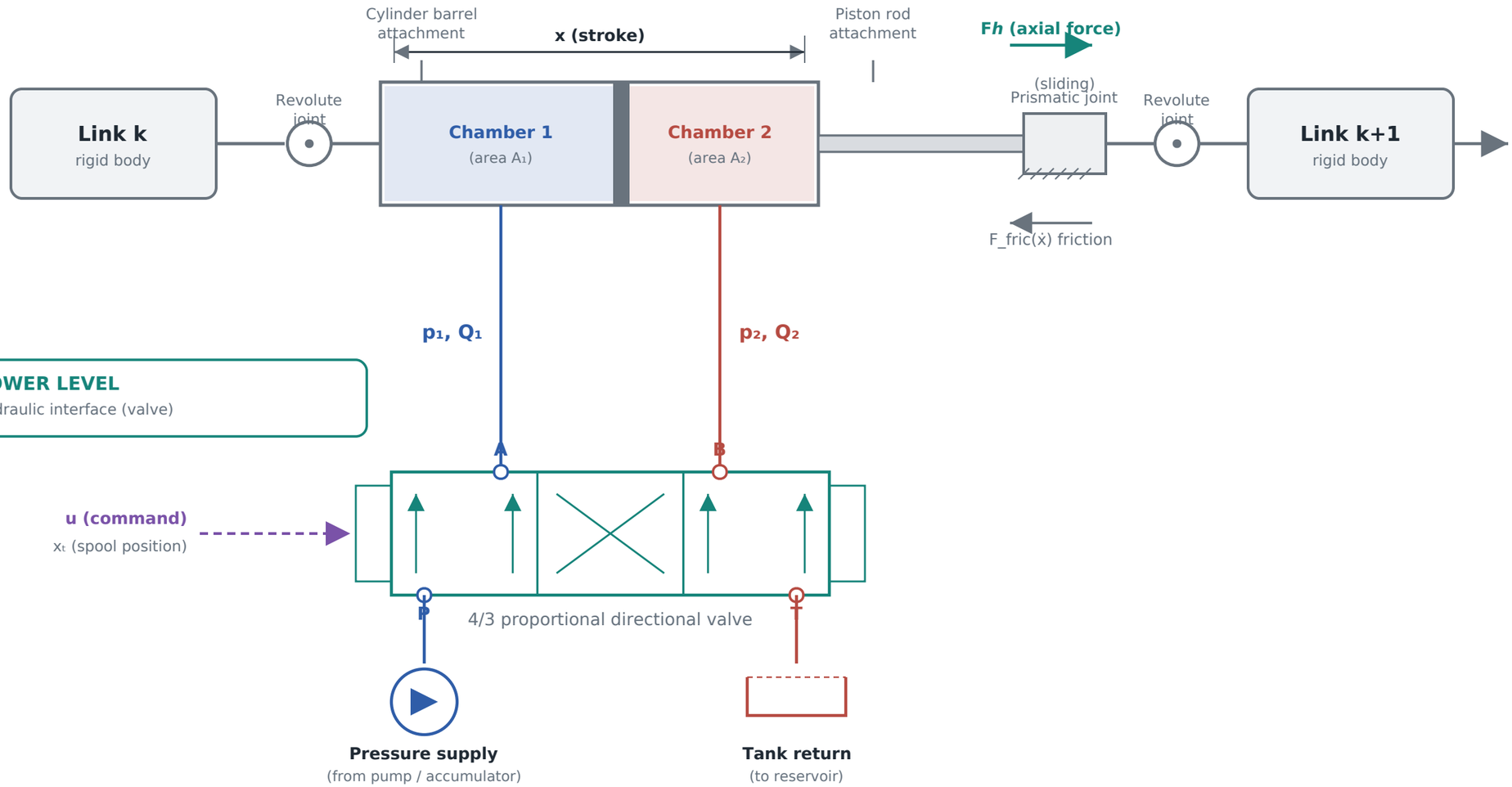
UPPER LEVEL
mechanical interface



LOWER LEVEL
hydraulic interface (valve)

LEGEND

- Mechanical connection
- Chamber 1 line (p_1,Q_1)
- Chamber 2 line (p_2,Q_2)
- Motion / force (Fh, F_fric)
- Command signal (u)

KEY VARIABLES

x, $\dot{x}$	cylinder stroke and velocity
p_1, p_2	chamber pressures
Q_1, Q_2	chamber flows
A_1, A_2	chamber (piston) areas
Fh	hydraulic force (axial)
$F_fric(\dot{x})$	friction force
F_l	external load
u	command to valve
x_t	valve spool position

Cylinder force

$$Fh = p_1A_1 - p_2A_2 - F_fric(\dot{x})$$

$\dot{x}_t = (u - x_t) / \tau_t$ (spool lag)